MARKET IMPACT
& ADOPTION



The Missing Layer:
Why Compute Hardware Needs a BOM Interchange Standard

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Meta | Infrastructure Supply Chain PLM

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Meta

What is this called?



What is this called?

- A: “Server”
- B: “Compute tray”
- C: “Compute sled”
- D: “Blade”
- E: “Server tray”
- F: “Host”
- G: “Node”
- H: Something else



What is this called?

YOU'RE RIGHT



What is a BOM?

A Bill of Materials (BOM) is a complete list of all the raw parts, items, components, and instructions needed to build, manufacture, or repair a product.

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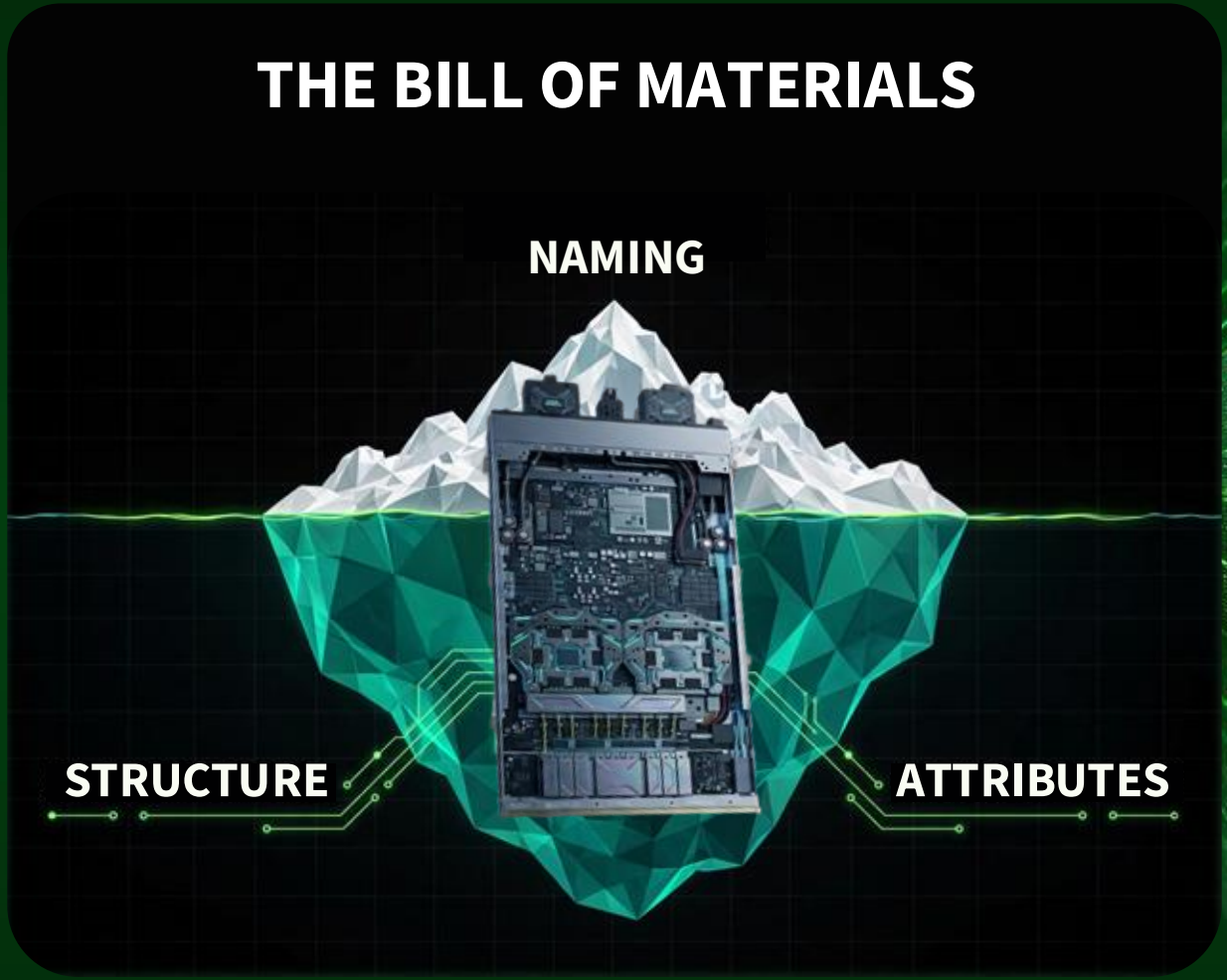
- *BOMs are the definitive recipes for products*

What is a BOM?

A Bill of Materials (BOM) is a complete list of all the raw parts, items, components, and instructions needed to build, manufacture, or repair a product.

- *BOMs are the definitive recipes for products*
- *BOMs are **exchanged** whenever product data needs to move in the supply chain*

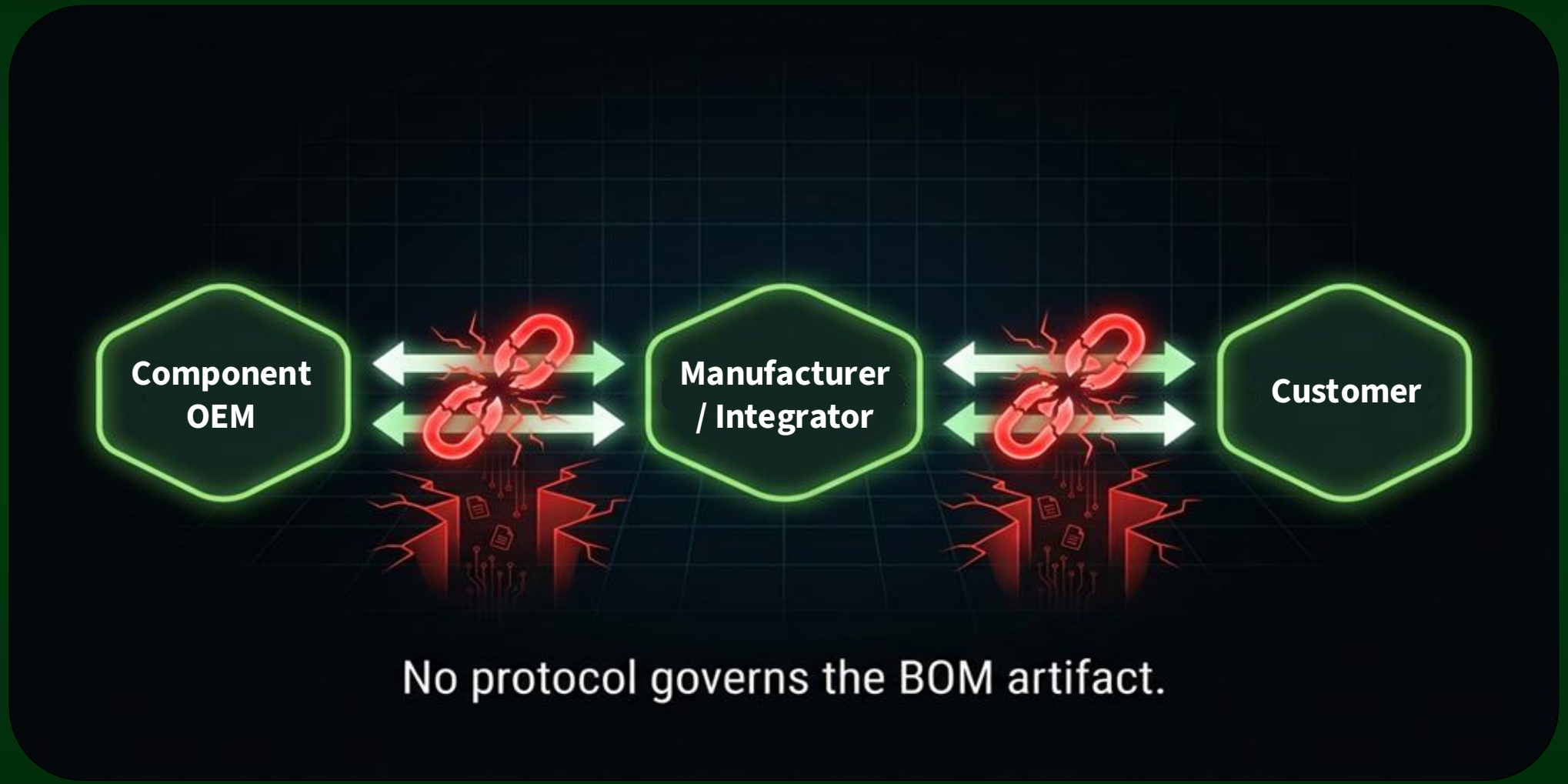
Naming Differences are only the Tip of the Iceberg



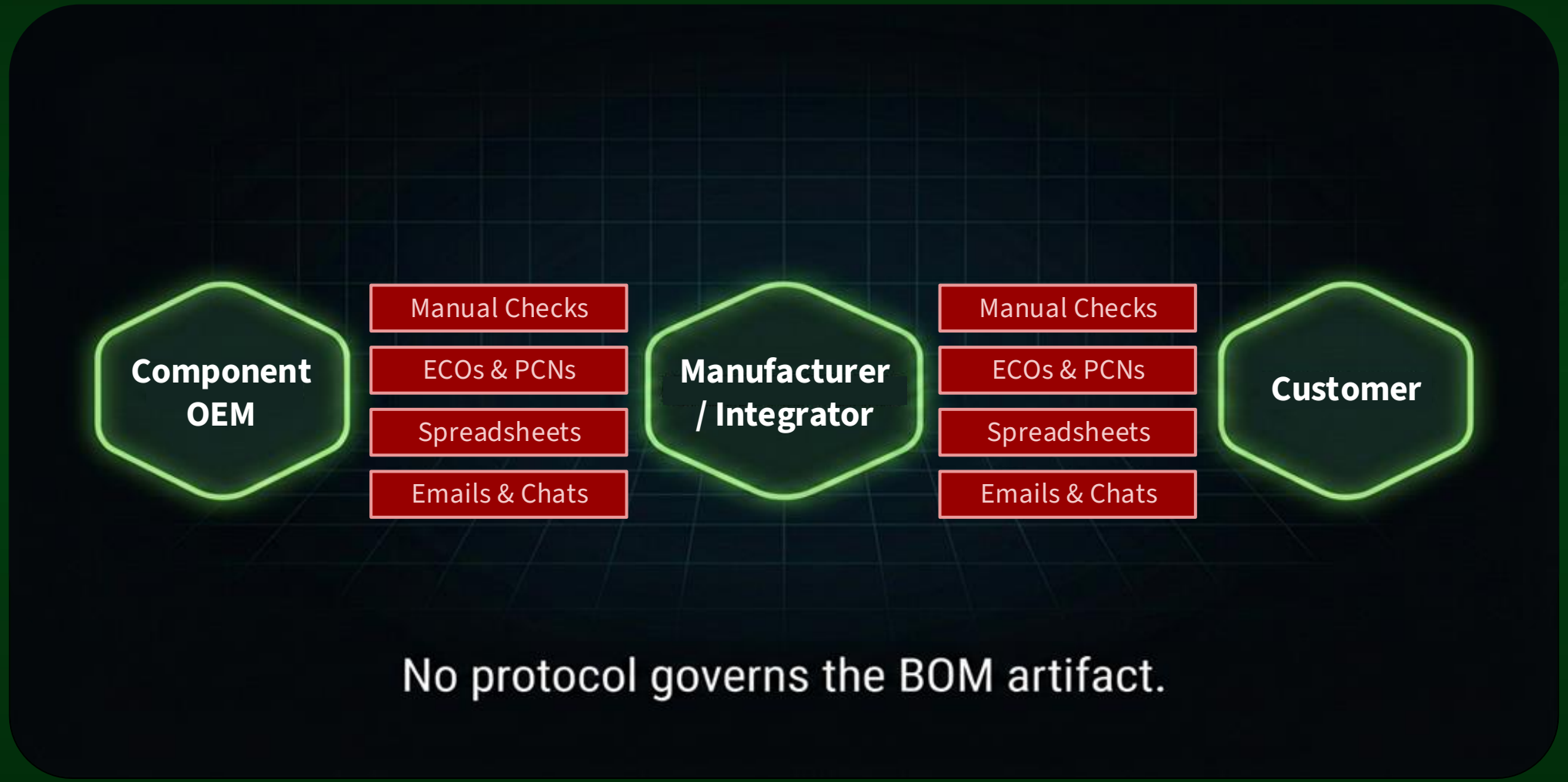
BOM Problems Live at *Interchange*



BOM Problems Live at *Interchange*



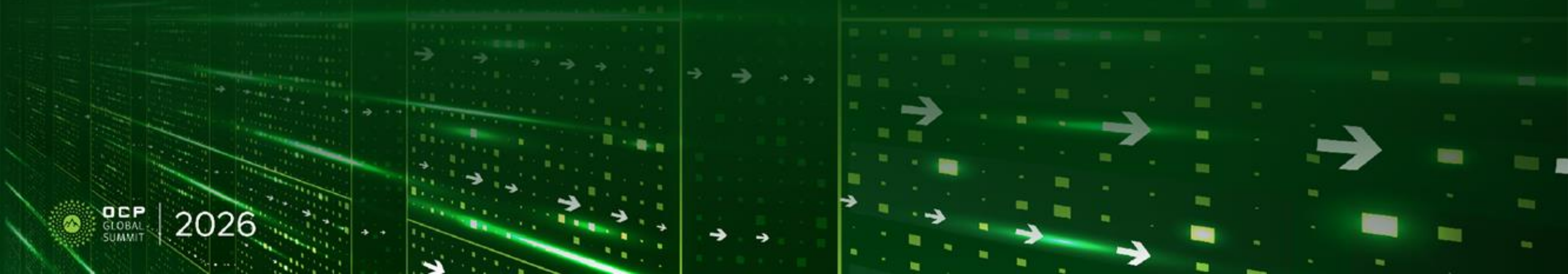
People and Processes Fill the Gaps



One Part. Left Behind.

Component OEM

Customer



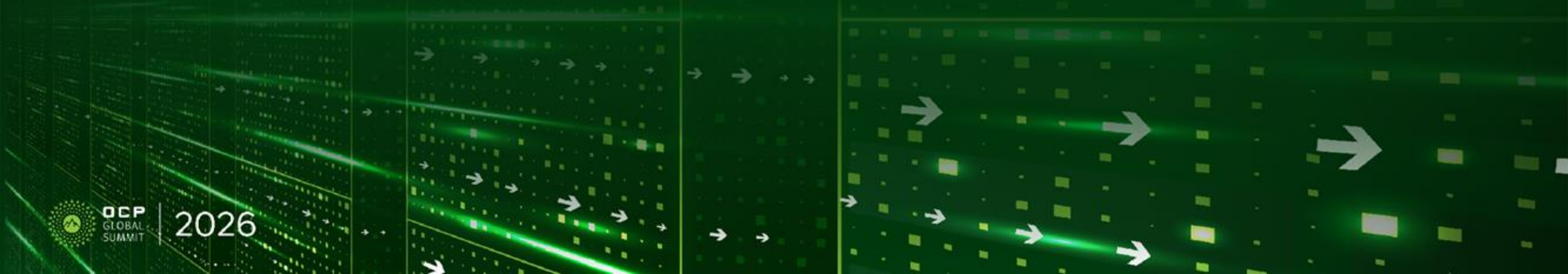
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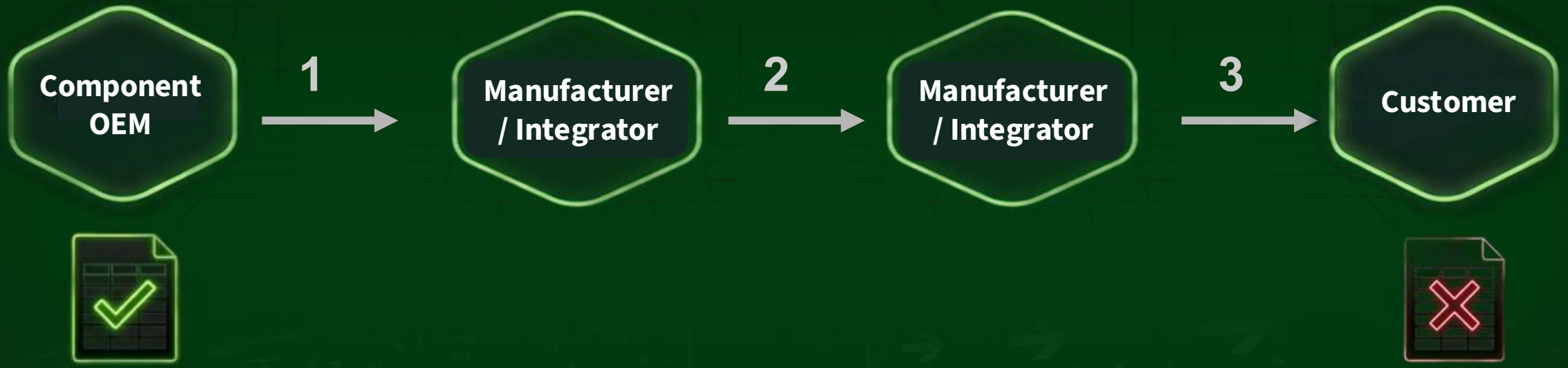
Manufacturer / Integrator

Manufacturer / Integrator

Customer



One Part. Left Behind.



What Does Interchange Cost?



Time, Money, and Quality

Interchange Cost = Fixed Cost + (Frequency x Lag x What it Blocks)

Fixed Cost

$/day

the current manual layer for PLM headcount, tooling, reconciliation

Frequency

67%

of BOMs are received with errors

Source: Supplyframe NPI Sourcing Study, 2022 (251 companies) intelligence.supplyframe.com

Lag

Days to Qtrs

how long until errors are noticed

What it Blocks

$/Day

POs, SLAs, MW Capacity, Cluster bringup time

Time, Money, and Quality

Interchange Cost = Fixed Cost + (Frequency x Lag x What it Blocks)

73%

of manufacturers cannot say what a change order costs them

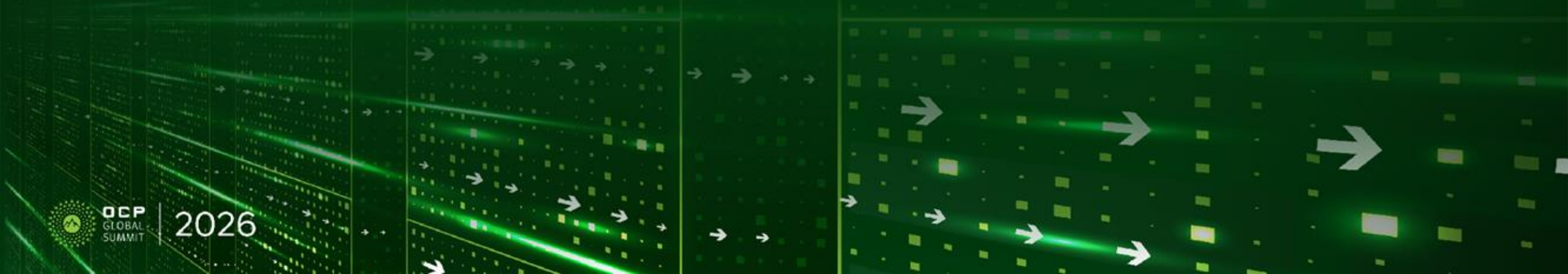
Cincom Systems · Best Practices: Mass Customization and Build-to-Order Manufacturing

Fixed Cost	Frequency	Lag	What It Blocks
?	?	?	?

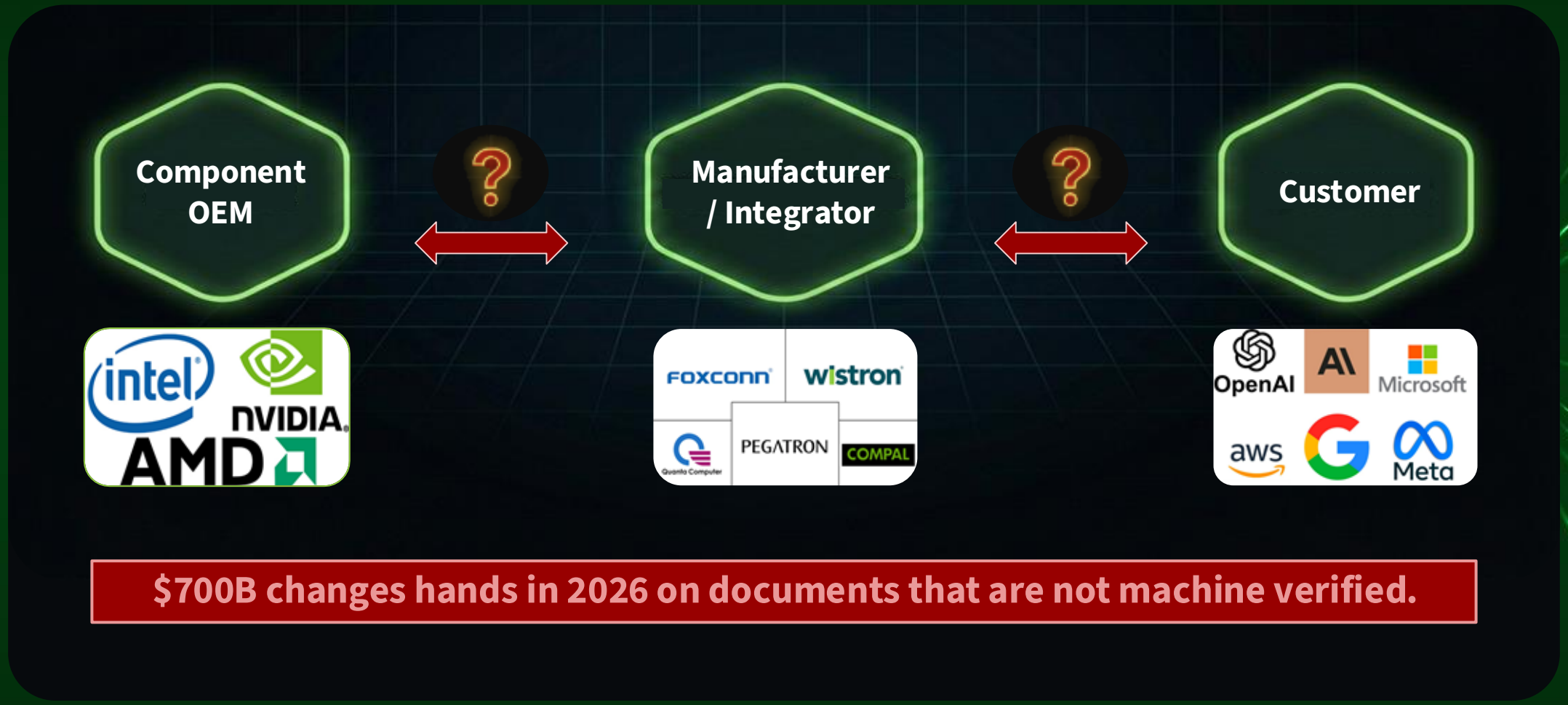
The Entire Supply Chain Pays

	COMPONENT OEMs	MANUFACTURERS / INTEGRATORS	CUSTOMERS / INFRA DEVELOPERS
TIME	Field engineering burned confirming which revision shipped	ECO and PCN cycles re-run on every revision	Rack and cluster bring-up slips
MONEY	Rebate, warranty and RMA claims adjudicated by hand	NRE, expedites, margin and rebates left unclaimed	Delay tax on stranded capacity and idle accelerators
QUALITY	Provenance unprovable - counterfeit and gray market compete	Built to a stale revision - rework, scrap, rejected changes	As-built differs from as-ordered; configs unprovable

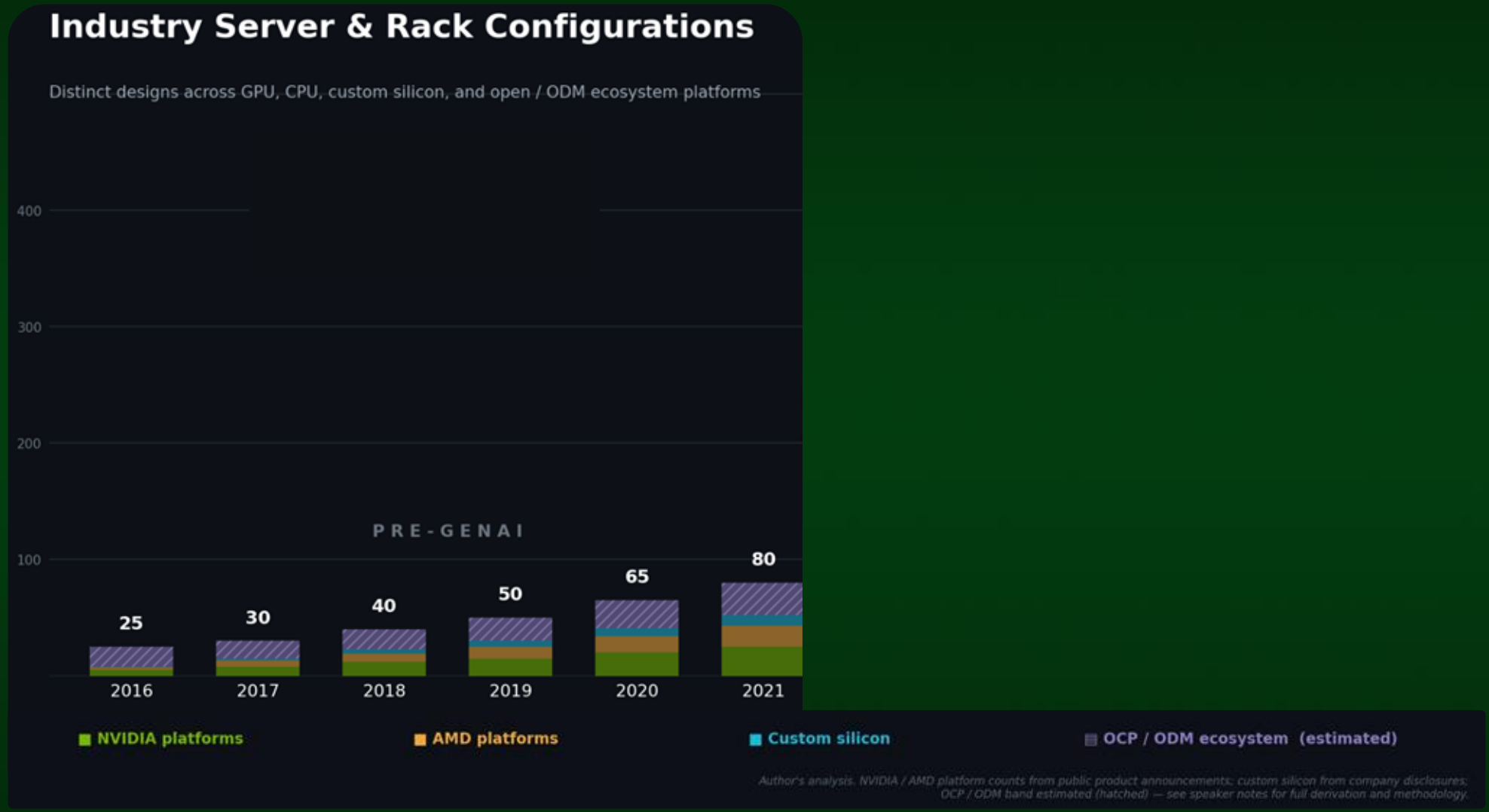
Why Solve BOM Interchange Now?



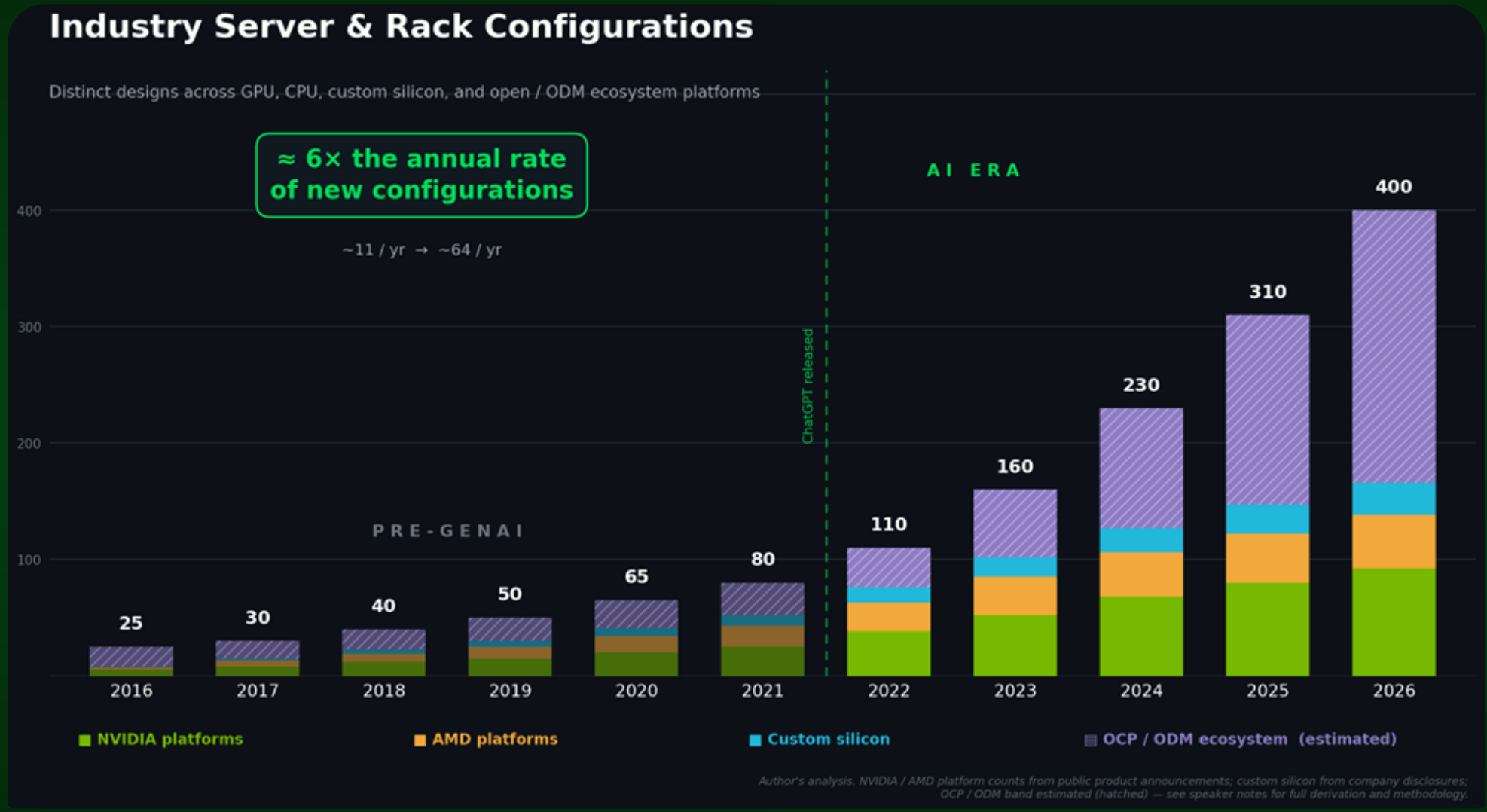
$700,000,000,000. No Common Receipt.



Configs are EXPLODING



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Policy is Being Written on an Unstandardized Layer



What do we do?

Build the Missing Layer.



Agree on Three Things



1. Naming

What is it called?

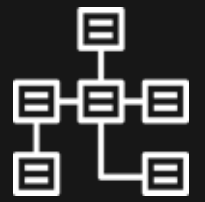
Universal taxonomy like software registries



2. Structure

How does it compose?

Standard BOM hierarchies like manifest files



3. Attributes

What must it carry?

Required data fields and values via type schema

Competitors Already Agree

	SOFTWARE *exists today*	HARDWARE *does not exist yet*
THE THREE AGREEMENTS		
Naming	`nginx:1.25`	`compute-tray:rev-C3`
Structure	Image manifest	BOM levels
Attributes	Required schema fields	Required component attributes
HOW THE AGREEMENTS TRAVEL		
Registry	Docker Hub	Addressable BOM store
Format spec	OCI Image Spec	BOM schema
Integrity	Sigstore	Signed BOMs

Extend, Don’t Invent

BOM Interchange Standard

Naming

JEDEC

GS1

Structure

CycloneDX

Attributes

CISA CYBER+INFRASTRUCTURE

IPC by Global Electronics Association

Version Control for Hardware

Before



Version Control for Hardware

Before



After

```
● 3 added | ● 1 removed | ● 47 qty changes | ● 2 structure moves

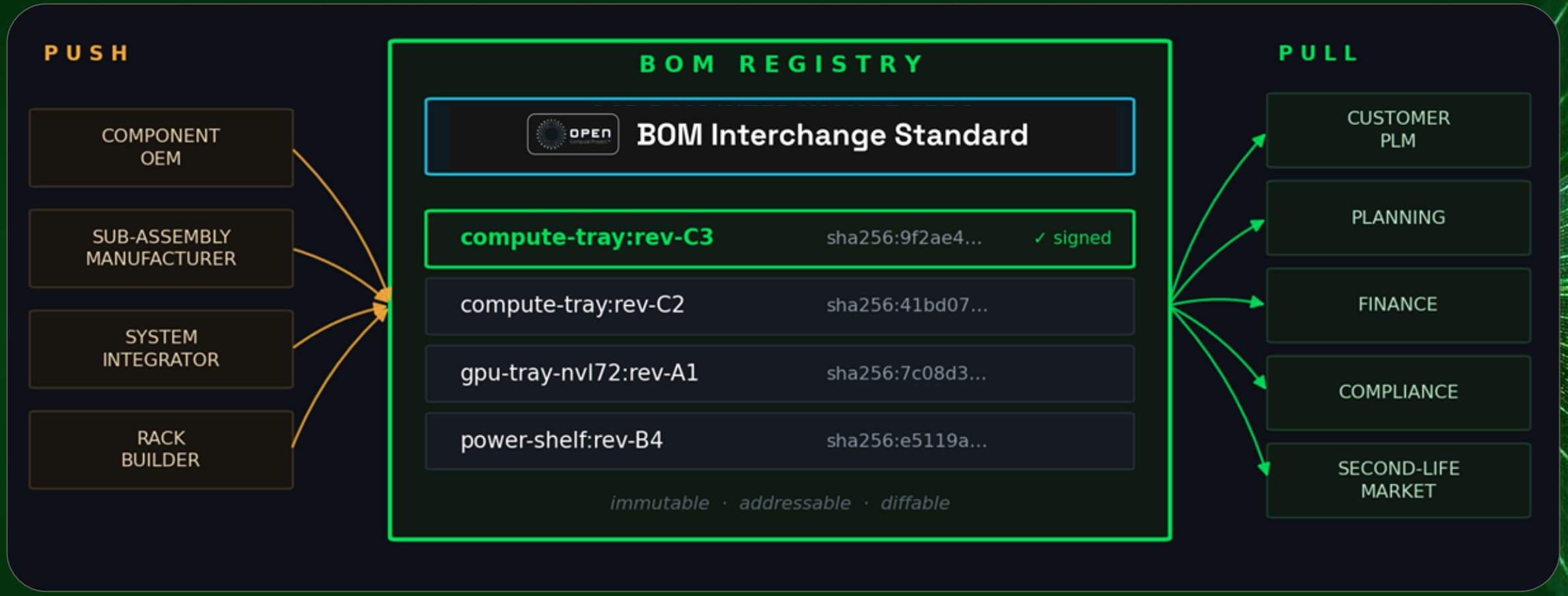
+ Samsung PM9D3 3.84TB NVMe U.2 SSD (MFR: MZWL23T8HCLS-00A07) × 8  /Chassis/Storage/Bay1-8
+ Broadcom BCM57504 25GbE NIC (MFR: BCM957504-P425G) × 2  /Chassis/Networking/PCIe-Slot1-2
+ Infineon XDPE192C3B VR Controller (MFR: XDPE192C3B) × 4  /Chassis/Motherboard/Power-Delivery

- Samsung PM9A3 1.92TB NVMe U.2 SSD (MFR: MZQL21T9HCJR-00A07) × 8  /Chassis/Storage/Bay1-8

~ SK Hynix HBM3E 36GB Stack (MFR: H3MRS236GA-B0RC): qty 32 → 48  /Chassis/GPU-Tray/Memory
~ Delta DPS-3000AB PSU 3000W (MFR: DPS-3000AB-26A): qty 2 → 4  /Chassis/Power-Bay/PSU1-4
~ Amphenol QSFP-DD 400G Cable (MFR: NDAAFF-0003): qty 16 → 24  /Chassis/Networking/Cabling

↳ Molex 74441-0010 Backplane Connector: moved from /Chassis/Midplane → /Chassis/GPU-Tray/Midplane
↳ TE 2357620-1 Power Distribution Board: moved from /Power → /Chassis/Power-Bay
```

The BOM Registry



Agree Once. Unblock Downstream.



Agree Once. Improve the Terms.

Interchange Cost = Fixed Cost + (Frequency x Lag x What it Blocks)

Fixed Cost

↓

Falls over time

Removes manual reconciliation required for data sharing today

Frequency

↓

Immediate Validation

Malformed data stops being transmissible. Ensures conformance, not judgement

Lag

↓

Seconds

Computed at ingest, not reactionary downstream

What it Blocks

Unchanged

A standard does not make capacity cheaper or change your POs

“But What About…”

- “Changing my tools?”
- “Exposing my margins?”
- “Revealing my design IP?”
- “My tree ≠ your tree?”

“But What About…”

- ~~“Changing my tools?”~~ -> This is a format, keep your tools.
- “Exposing my margins?”
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“But What About…”

~~“Changing my tools?”~~ -> This is a format, keep your tools.

~~“Exposing my margins?”~~ -> Schema carries parts, not prices.

“Revealing my design IP?”

“My tree ≠ your tree?”

“But What About…”

~~“Changing my tools?”~~	->	This is a format, keep your tools.
~~“Exposing my margins?”~~	->	Schema carries parts, not prices.
~~“Revealing my design IP?”~~	->	Ship images. Not Dockerfiles.

“My tree ≠ your tree?”

“But What About…”

~~“Changing my tools?”~~	->	This is a format, keep your tools.
~~“Exposing my margins?”~~	->	Schema carries parts, not prices.
~~“Revealing my design IP?”~~	->	Ship images. Not Dockerfiles.
~~“My tree ≠ your tree?”~~	->	Different trees. Now computable.

Join the OCP BOM Interchange Workstream

Deliverables

1. Universal BOM Definition & Types
1. Attribute Schema, Shapes & Repo
1. Pilot Exchange
1. Published Spec

Timeline

- Q4 2026 - Charter & Co-leads appointed
- Q1-2 2027 - Draft taxonomy & schema
- Q3 2027 - Pilot
- Q4 2027 - Publish v1.0 via OCP

Participation

Committed Partners:

- Meta
- Quanta
- Ingrasys (Foxconn)

We Still Need:

- **Co-Leads**
- SW & Data Leaders
- OEM PLM Leaders
- Hyperscale PLM Leaders
- Neocloud PLM Leaders

Be in the Room



OCP GLOBAL SUMMIT | 2026



Looking for co-leads and PLM leaders.
Reply directly or scan:

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wrightjacob@meta.com